ForceDelta-VLA: Supplementary Material

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I. Training and Architecture Details

The teacher is trained for 50,000 steps with AdamW at a peak learning rate of 2×10^{-5}. It predicts H=50 actions at 30 Hz from the latest 100 ms of the 100 Hz wrench stream, and flow-matching inference uses N_{flow}=10 steps.

After freezing the teacher, we train the rank-32 missing-force adapter for 10,000 steps at a learning rate of 10^{-4}. The visual–language prefix is pooled into M=16 positional bins and projected to Q=2 intent tokens. The residual policy comprises a causal force encoder and one pre-norm set-attention block with separate zero-initialized force and staleness output projections. It predicts K=5 corrections and is trained for 30,000 steps at a learning rate of 3×10^{-4} under the sampled asynchronous schedule. The staleness pass masks the force-token position in the attention-validity mask.

The action representation and rotational residual composition are defined in Sec. III-A of the main paper. Let $\widetilde{S}^p$ be the first nine pose coordinates of the normalized robot state, obtained after converting orientation to the continuous 6D representation, and let σ_S^p and σ_A^p be the corresponding training-set state and action standard deviations. The implementation computes

$$\Gamma(S_t, S_k) = \frac{\sigma_S^p + \varepsilon}{\sigma_A^p + \varepsilon} \odot (\widetilde{S}_t^p - \widetilde{S}_k^p), \qquad \varepsilon = 10^{-6}. \quad (1)$$

This scale-corrected componentwise difference is included in the learned staleness target, placing the current missing-force prediction and cached reference in the same normalized delta-action coordinate system. The composed normalized action is de-normalized, applied relative to the reference-query state S_k, and projected to a valid rotation by Gram–Schmidt orthogonalization.

The residual policy observes the complete K-step reference segment aligned with its output timestamps. The segment is flattened and projected to one typed reference token, so each predicted correction is conditioned on the corresponding future reference waypoint without increasing the attention-sequence length.

At deployment, force-conditioned and staleness corrections are clipped separately and element-wise before composition. The implementation accepts either nine-dimensional nonnegative bounds $b_F, b_S \in \mathbb{R}^9_{\geq 0}$ or scalar values broadcast over the pose coordinates; bimanual clipping is applied independently per arm. The final robot configuration must specify the values used for b_F and b_S.

Residual-policy training examples are anchored at the 30 Hz state/action timestamps while retaining every 100 Hz wrench sample in $\mathcal{F}_t$. The sampled schedule uses inter-query periods $T_i \sim \mathcal{U}(1/15, 1/5)$ s and reference-query latencies $L_i \sim \mathcal{U}(0.05, 0.30)$ s. Query rows are first selected on the recorded grid, after which query timestamps receive $J_i \sim \mathcal{U}(-0.01, 0.01)$ s jitter to exercise off-grid reference interpolation. Residual queries may start more frequently at deployment; the realized correction-update rate is measured directly rather than inferred from the action-sample rate within a chunk.

A. Missing-Force Mode Optimization

The learned missing-force token and mode-specific adapter are active only for missing-force queries. With the force-conditioned teacher parameters θ fixed, the missing-force parameters ϕ are optimized using the teacher's original flow-matching objective,

$$\mathcal{L}_{\text{miss}} = \mathbb{E}_{\mathcal{D},\eta,\epsilon}\left[\ell_{\text{FM}}\left(T_{\theta,\phi}^{\text{miss}}, A^*_{t:t+H-1}; \eta, \epsilon\right)\right], \quad (2)$$

where η is flow time and ϵ is the initial flow noise. Force-conditioned and missing-force predictions used to construct a residual target share the same ϵ.

B. Residual-Loss Weighting

For a K-step residual chunk, the normalized temporal weights are

$$w_j = \frac{\beta^{j-1}}{\sum_{m=1}^{K} \beta^{m-1}}, \qquad 0 < \beta \leq 1. \quad (3)$$

Thus, $\beta < 1$ emphasizes near-term corrections, $\beta = 1$ weights all steps equally, and $\sum_j w_j = 1$ keeps the loss scale stable as K changes. The loss is a mean-squared error in normalized pose-action coordinates.

II. Asynchronous Reference and Correction Execution

A. Reference Updates

A reference query k, started at t_k from state S_k, publishes the immutable reference update $(A^{\text{miss}}_{T,k,1:H}, Z^{\text{intent}}_k, S_k, t_k, \bar{t}_k)$ when its result becomes available at $\bar{t}_k$. Its samples carry intended execution times

$$t^{\text{ref}}_{k,n} = t_k + (n-1)/f_A, \qquad n = 1, \ldots, H. \quad (4)$$

At time t, the active reference is the freshest completed, non-superseded query,

$$k^*(t) = \arg\max_{k:\, \bar{t}_k \leq t} t_k. \quad (5)$$

An older query that completes later therefore cannot replace a newer active reference update. Samples whose intended times precede $\bar{t}_{k^*}$ are skipped for direct execution; the cached chunk itself is not truncated. For $t \geq \bar{t}_{k^*}$, the reference pose $A^{\mathrm{ref}}_{k^*}(t)$ is obtained by timestamp-based interpolation over the original timestamped chunk.

B. Correction Updates

A residual query q, started at t^F_q, reads $k_q = k^*(t^F_q)$ together with $(\mathcal{F}_{t^F_q}, S_{t^F_q})$. It completes at $\bar{t}^F_q$ with force-conditioned and staleness correction chunks. Their samples are timestamped as

$$t^F_{q,j} = t^F_q + (j-1)/f_A, \qquad j = 1, \ldots, K. \tag{6}$$

Here f_A defines the spacing between correction samples within a chunk; residual query start times t^F_q are scheduled independently. Step j is held over $[t^F_{q,j}, t^F_{q,j} + 1/f_A)$, and correction samples are not interpolated. At control time t, the executor selects the most recently started residual query whose result has completed and remains valid, together with its active step:

$$\begin{aligned} q^*(t) &= \arg\max_{q:\, \bar{t}^F_q \le t < t^F_q + K/f_A} t^F_q, \\ j(t; q^*) &= 1 + \left\lfloor (t - t^F_{q^*}) f_A \right\rfloor. \end{aligned} \tag{7}$$

Any prefix that expires during inference is therefore skipped, while the remaining valid steps execute sequentially until a newer correction chunk completes. Let $j^* = j(t; q^*)$. Within the selected interval, Eq. (1) of the main paper is evaluated using the selected residual query time and active step:

$$\begin{aligned} \mathcal{P}(A^{\mathrm{cmd}}(t)) &\equiv \mathcal{P}(A^{\mathrm{cmd}}_{t^F_{q^*}, j^*}) \\ &= \mathcal{P}\Big(A^{\mathrm{ref}}_{k_{q^*}}(t^F_{q^*,j^*})\Big) + \Delta\hat{A}^{\mathrm{force}}_{q^*,j^*} + \Delta\hat{A}^{\mathrm{stale}}_{q^*,j^*}. \end{aligned} \tag{8}$$

The two corrections are clipped independently and element-wise in normalized delta-action coordinates before composition. The composed command remains referenced to $S_{k_{q^*}}$ and is converted to an absolute robot command as described above. A correction chunk that completes after $t^F_q + K/f_A$ is rejected. If no valid correction chunk remains, the executor invokes the platform's safe hold or abort policy. The final implementation record will specify gripper sampling and behavior beyond the last valid reference timestamp.

III. Additional Evaluation Protocols

A. Task Success Criteria

The task-specific success criteria are summarized in Table I.

B. Metric Aggregation

Peak force is computed from every trial. For a bimanual trial, it is the maximum force norm over time and over both arms. Completion time is measured only on successful trials, from the first policy command to satisfaction of the task-specific success criterion. Platform-level values first aggregate trials within each task and then macro-average the five single-arm or four bimanual tasks. These conventions must be verified against the final evaluation code.

C. Task-Wise Trial Counts

The final experiment record will report the successful-trial count for each method and task. Each count is out of twenty evaluation trials and aggregates to the main paper's success-rate table.

D. Residual-Bound Sensitivity

After fixing the robot-deployment bounds (b_F, b_S), we jointly scale them by $0.5\times$, $1\times$, and $2\times$ on USB insertion and drawer opening. The two heads remain clipped separately and element-wise, with independent per-arm clipping in the bimanual setting (Table II).

TABLE I
Task success criteria. Red entries are provisional thresholds.

Platform	Task	Success criterion
Single-arm	Object Flipping	Object rotated by at least 150° and left stable
Single-arm	USB Insertion	Connector seated to within 2 mm of its terminal depth
Single-arm	Cabinet Opening	Door opened beyond 60°
Single-arm	Button Pressing	Button fully actuated
Single-arm	Whiteboard Wiping	At least 90% of the marked region removed
Bimanual	Plug Removal	Plug fully separated without dropping either component
Bimanual	Plug Insertion	Plug fully seated while the second arm stabilizes the socket
Bimanual	Drawer Opening	Drawer translated by at least 15 cm
Bimanual	Whiteboard Wiping	At least 90% of the marked region removed while the left arm stabilizes the board and the right arm wipes

TABLE II
Sensitivity to the residual bound. Red entries are placeholders.

	USB insertion		Drawer opening	
Bound	Success (%)	Peak (N)	Success (%)	Peak (N)
0.5×	X	X	X	X
1×	X	X	X	X
2×	X	X	X	X